



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

AGRICULTURAL TECHNOLOGY

2022

MARKING GUIDELINES

MARKS: 200

These marking guidelines consist of 15 pages.

SECTION A**QUESTION 1**

1.1 1.1.1 A✓✓

1.1.2 C✓✓

1.1.3 D✓✓

1.1.4 B✓✓

1.1.5 A✓✓

1.1.6 C✓✓

1.1.7 B✓✓

1.1.8 D✓✓

1.1.9 C✓✓

1.1.10 A✓✓

(20)

1.2 1.2.1 Red ✓✓

1.2.2 Hard facing ✓✓

1.2.3 Noise pollution ✓✓

1.2.4 GPS ✓✓

1.2.5 Rectangular bales ✓✓

(10)

1.3 1.3.1 E✓✓

1.3.2 G✓✓

1.3.3 C✓✓

1.3.4 F✓✓

1.3.5 D✓✓

(10)

TOTAL SECTION A: 40

SECTION B**QUESTION 2: MATERIALS AND STRUCTURES**

- 2.1 THREE possible factors that when choosing a suitable adhesive for this specific application.
- Inflammability.✓
 - Duration of cohesion.✓
 - Duration of usability.✓
 - Heat resistance. (temperature)✓
 - Water resistance/ Oil resistance.✓
 - Elasticity.✓
 - Load capacity.✓
- (Any 3) (3)
- 2.2 Advantages of using a water trough made from glass fibre over a trough made of steel.
- Lightness.✓
 - Can be formed into any shape.✓
 - Can easily be sawn, drilled, and filed.✓
 - Toughness.✓
 - Easy repaired when broken.✓
 - Does not rust, corrode or erode. ✓
- (Any 3) (3)
- 2.3 FOUR chemical substances that does not have any effect on Teflon.
- Adhesives./ Glue.✓
 - Asphalt/tar.✓
 - Dyes./ Poisons✓
 - Greases.✓
 - Gasses. ✓
 - Latex.✓
 - Lacquers/ Fuels.✓
 - Paint.✓
 - Acids. ✓
- (Any 4) (4)
- 2.4 2.4.1 ONE reason for using Vesconite in dry applications and explanation for each answer.
- Reason: Vesconite do not need any lubricant. ✓
- Explanation: Made of internal lubricated polymers. ✓
- (2)
- 2.4.2 TWO circumstances where Vesconite can be effectively used on a tractor to replace existing metal bushes.
- Front-axle swivel bushes.✓
 - Steering linkage bushes.✓
 - Gear lever bushes. ✓
 - Three point lift bushes. ✓
- (2)

- 2.5 2.5.1 THREE factors that must be taken into consideration when identifying tin for the manufacturing of food cans.
- Soft.✓
 - Malleable metal.✓
 - Can be highly polished.✓
 - Resists oxygen and water but dissolves in acids and bases (prevent rust).✓
- (Any 3) (3)
- 2.5.2 TWO commercial uses of tin other than the application in the food canning industry.
- Metal coating.✓
 - Alloy element of bronze.✓
 - Alloy element of soft soldering.✓
 - Cool drink cans. ✓
- (Any 2) (2)
- 2.6 TWO properties of bronze bushes that makes it better suited for the use on implements.
- Bronze resists corrosion.✓
 - Resists metal fatigue more than steel.✓
 - Better conductor of heat.✓
 - Low friction properties.✓
- (Any 2) (2)
- 2.7 THREE influences of manganese on stainless steel.
- It combats corrosion.✓
 - Gives steel a coarser structure.✓
 - Changes the band structure, causing a reduction in striking strength.✓
 - Increases tensile strength.✓
 - Reduces the critical cooling tempo.✓
 - Improves hardening.✓
 - Increases resistance against wear.✓
 - Reduces magnetism.✓
- (Any 3) (3)
- 2.8 Justification of the use of brass over copper in the manufacturing of water couplings.
- Strength.✓
 - Machinability.✓
 - Wear resistance.✓
 - Hardness.✓
- (Any 2) (2)

- 2.9 2.9.1 Identify component **A and B** in the diagram.
- A - Energiser.✓
 B - Earth spike.✓ (2)
- 2.9.2 The maximum voltage allowed by legislation that can be applied in the system.
- 10 000 volt.✓ (1)
- 2.9.3 Description of the daily tasks that must be carried out to maintain an electric fence.
- Clear any vegetation from the fence line.✓
 - Check for damaged or broken wires.✓
 - Damaged isolators or loose connections.✓
 - Regular testing of the energiser's pulse.✓ (Any 2) (2)
- 2.9.4 FOUR soil conditions that can have a negative effect on the earth efficiency of an electrical fence.
- Peat.✓
 - Sandy soil.✓
 - Gravel.✓
 - Very dry soil.✓
 - Snow or frozen ground.✓ (Any 4) (4)
- [35]**

QUESTION 3: ENERGY

3.1 3.1.1 THREE important factors that must be considered when installing a wind turbine.

- Select suitable environment.✓
- Size of the turbine.✓
- Availability of substantial wind strength.✓
- Absence of mountains and hills.✓
- Turbine capacity.✓
- Hire a professional to do a survey on the surrounding area.✓ (Any 3) (3)

3.1.2 Explanation of TWO benefits of wind turbines.

- Decades of free electricity after initial-cost recovery.✓
- Increased property values.✓
- Reliable electricity generation.✓
- Relief from high prices of other forms of electricity.✓
- Personal energy independence.✓
- Supports clean energy.✓
- Fight global warming.✓
- Renewable energy. ✓
- No fuel costs. ✓ (Any 2) (2)

3.2 3.2.1 The semi-conductive material used for the manufacturing of the photovoltaic solar panel.

Silicon.✓ (1)

3.2.2 Explanation of the process of generating electrical energy in a solar panel.

- The solar panels are made of a semi-conductive material that contains electrons.✓
- When photons (contained within the sun's rays) hit the solar cells, the electrons absorb this solar energy.✓
- Transforming them into conduction electrons.✓
- Electrons are able to become free, and carry an electric charge through a circuit to a destination.✓ (4)

3.3 The advantages of a geothermal power station above a coal power station.

- A geothermal system does not create any pollution.✓
- The cost of the land to build a geothermal power plant on is usually less expensive.✓
- Geothermal plants take up very little room.✓
- You may receive tax cuts, and/or no environmental bills.✓
- No fuel is used to generate the power.✓
- No costs for purchasing, transporting, or cleaning up of fuels.✓ (Any 4) (4)

3.4 FOUR benefits of biofuels.

- Biofuel offers a cheaper solution to our energy needs.✓
- Bio-fuels are made from plant and animal waste.✓
- Biodegradable.✓
- Do not harm the environment.✓
- Does not require any radical changes to switch to the use of biofuels.✓
- Renewable sources of energy.✓
- Inexpensive to produce.✓
- Help prevent engine knocking.✓

(Any 4)

(4)

3.5 TWO resources for the manufacturing of methanol.

- Woody plant fibre.✓
- Coal.✓
- Natural gas.✓
- Fermented waste products such as sewage and manure.✓

(Any 2)

(2)

[20]

QUESTION 4: SKILLS AND CONSTRUCTION PROCESSES

- 4.1 4.1.1 Identification of part A.
- Cutting Nozzle.✓ (1)
- 4.1.2 Identifying the problem indicated by arrow B that can occur when thick materials are being cut with the plasma-cutting machine.
- The angle of cut will not be square/90°.✓ (1)
- 4.1.3 Addressing the problem identified in QUESTION 4.1.2.
- A machine can be used to grind the face square.✓
 - The welding nozzle can be tilted at an angle to compensate for the problem.✓ (Any 1) (1)
- 4.1.4 TWO types of gasses that are commonly used in the plasma-cutting process.
- Regular air✓
 - Argon✓
 - Nitrogen✓
 - Oxygen✓ (Any 2) (2)
- 4.2 4.2.1 TWO gasses used during the oxy-acetylene cutting process.
- Acetylene✓ and oxygen. ✓ (2)
- 4.2.2 Advantages of the oxy-acetylene apparatus over the plasma cutter.
- No need for electricity.✓
 - Can be used to heat up work pieces. ✓
 - Portable.✓
 - No electrical components.✓
 - Rust has no influence on the cutting process.✓
 - Easy to operate. ✓ (Any 3) (3)

4.2.3 Important safety measures to note when working with the oxy-acetylene cutting apparatus.

- If a cylinder falls over and breaks the main valve off, the cylinder will become a missile and cause extreme damage.✓
- Wear a leather apron or similar protective clothing and welding gloves when using an oxy-acetylene cutting torch.✓
- Always use proper oxy-acetylene cutting goggles.✓
- Never point the flame towards another person or any flammable material.✓
- Always light the oxyacetylene cutting torch with a striker.✓
- Wherever possible, use a heat shield behind the component you are heating.✓
- After heating a piece of metal, label it as 'HOT' with a piece of chalk so that others will not attempt to pick it up.✓
- Make sure there are no leaks on pipes and connections. ✓
- Make sure all valves are closed after use. ✓

(3)

(Any 3)

4.3 4.3.1 Describing the process to replace a worn welding tip.

- Remove the welding shield cup.✓
- Unscrew the damaged welding tip.✓
- Screw the new tip on.✓
- Replace the welding cup.✓

(4)

4.3.2 Explanation of the use of anti-spatter spray during the MIG welding process.

- Prevent the sprout from clogging with welding metal.✓
- Prevent the filler wire/welding electrode from sticking to the contact tip.✓

(2)

4.3.3 FOUR reasons for the welding wire not running smoothly through the welding hose.

- Bended welding feeder hose.✓
- Damage to the feeding mechanism.✓
- Corroded welding electrode/wire.✓
- Damaged tip.✓

(4)

4.4 Preventative measures:

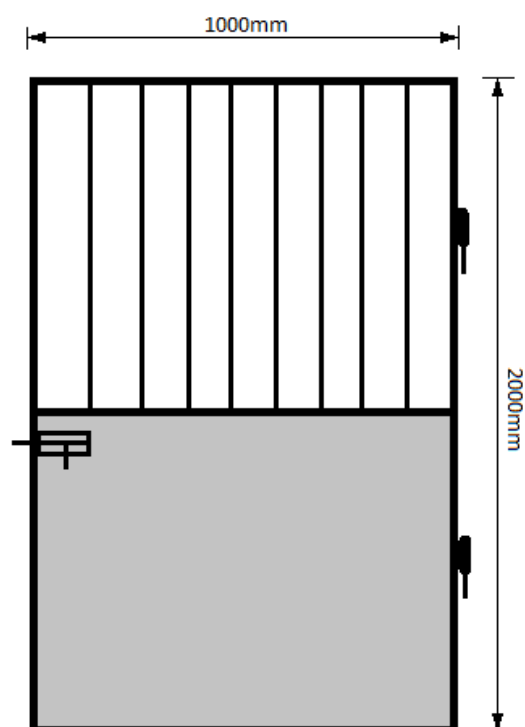
- 4.4.1 Spot weld.✓
- 4.4.2 Pre-setting.✓
- 4.4.3 Clamping.✓

(3)

4.5 Design drawing of a door for a horse stable.

Marks will be allocated for the following:

Design	(1)✓
Hinges and latch	(2)✓✓
Dimensions	(2)✓✓
Neatness	(1)✓



(6)

4.6 Explanation of using an inverter welder powered by a generator rather than using a MIG welder.

- Lightweight.✓
- Compact.✓
- No need for gas cylinder.✓
- Inverter can work in windy conditions.✓

(Any 3)

(3)
[35]

QUESTION 5: TOOLS, IMPLEMENTS AND EQUIPMENT

5.1 5.1.1 FIVE basic implements that can be used in the harvesting of the crop.

- Tractor.✓
 - Cutting machine.✓
 - Hay rake.✓
 - Baling machine.✓
 - Wrapping machine.✓
 - Front-end loader.✓
- (Any 5) (5)

5.1.2 THREE advantages of using machinery in the harvesting process.

- Single operation.✓
 - Less time consuming.✓
 - Labour saving.✓
 - Very reliable method.✓
 - Economical.✓
- (3)

5.1.3 Another method that can be used to cut lucerne.

Using a sickle.✓ (1)

5.1.4 ONE safety device that is installed on a baling machine

- Shear bolt.✓
 - Slip clutch.✓
 - Tension springs.✓
- (Any 1) (1)

5.2 5.2.1 Calculation of the running cost of a combine harvester.

(Show ALL calculations.)

R8 100-00 + R1 200-00 + R1 500-00 + R3 000-00
+ R8 000-00 = R21 800-00✓✓ (2)

5.2.2 Calculating the VAT (15%) of the total running cost.

R21 800-00 x 15% = R3 270-00✓✓ (2)

5.3 5.3.1 Label for the graph.

Depreciation.✓ (1)

5.3.2 Determine the right time to sell the tractor by analysing the data from the graph and TWO reasons for the answer.

- During year 4.✓
- **Motivation**
- One year left on maintenance plan.✓
- Higher value than selling during year 6.✓ (3)

5.3.3 THREE actions a farmer can take to minimize excessive depreciation in the value of second-hand implements.

- Repair/Replace broken or worn parts.✓
- Store properly.✓
- Service according to user manual.✓
- Use implement according to specifications.✓ (Any 3) (3)

5.4 5.4.1 Identification of the components labelled **A**, **B** and **C** and TWO functions of each.

A. Top link.✓

- To adjust the angle of the implement in relation to the tractors movement.✓
- Serves as top connection of the three-point mechanism to the implement.✓

B. Hydraulic pump.✓

- Provides pressure to the hydraulic system of the tractor.✓
- Provides pressure to the hydraulic system of the implement.✓

C. PTO drive shaft.✓

- Transmits driving power from tractor to the implement.✓
- Provide angular movement between the tractor and implement.✓ (9)

5.4.2 THREE important safety precautions applicable to component **C**.

- Never climb over the driving shaft when in motion. ✓
- Safety screen must be in place.✓
- Safety screen must not rotate with the shaft.✓
- Screen must be highly visible.✓
- Never work on an implement when driving shaft is in motion.✓ (Any 3) (3)

5.5 5.5.1 Identification of the components labelled **A**, **B** and **C**.

A	Flywheel.✓
B	Clutch plate.✓
C	Pressure plate.✓

(3)

5.5.2 FOUR reasons for equipping a tractor with a clutch.

- Engine drive needs to be disengaged when gears are changed.✓
- Drive should be disengaged when the tractor is started.✓
- The clutch is disengaged to allow engine speed to increase and then engaged to give greater torque.✓
- Allows the operator to stop the tractor, belt pulley or PTO shaft without stopping the engine.✓

(4)

[40]

QUESTION 6: WATER MANAGEMENT

- 6.1 6.1.1 Explanation of the structure's ability to carry the heavy load of the irrigation system.
- Sustained by triangulation/truss method.✓
 - Arched design.✓
 - Steel cables/rods hold the trusses.✓ (Any 2) (2)
- 6.1.2 ONE possible cause of sprinkler nozzle blockage and provide a solution to the problem.
- Sand/fertilizer particles.✓
 - Plant matter.✓
 - Organisms in water. (Algae, tadpoles etc.)✓
 - This can be prevented by installing a filter in the system.✓ (Any 2) (2)
- 6.1.3 THREE factors to consider when selecting a water pump for the centre pivot system.
- Water source.✓
 - Type of pump.✓
 - Size.✓
 - Pressure requirements.✓
 - Available pump models.✓
 - Power source needed to run the pump.✓ (Any 3) (3)
- 6.2 The process when an irrigation farmer sets the correct frequency and duration of water application to a crop to maximise plant growth.
- Irrigation scheduling/timing.✓ (1)
- 6.3 6.3.1 Identify component **A** and its function.
- Manhole/Drain cover.✓
 - It is to provide access for cleaning and inspection.✓ (2)
- 6.3.2 Requirements that must be followed to keep septic system functional.
- Use only toilet paper.✓
 - Do not flush down non-degradable materials into the tank. (Cigarettes, plastics, rubber)✓
 - No disinfectants, bleaches, oils should be flushed down the system.✓
 - Inspect manhole regularly.✓
 - Empty the tank when required.✓ (Any 4) (4)

- 6.4 6.4.1 Explanation of the technical lay out of the drainage system.
- It contains perforated pipes.✓
 - The pipes are buried under gravel or pebbles.✓
 - The water drains through the surface gravel and seeps into the pipe's perforations before flowing out at the end of the pipe.✓
 - Pipes are installed at a slight angle or slope to facilitate the flow of water away from the waterlogged area.✓ (Any 3) (3)
- 6.4.2 A system that can quickly move large amounts of water from water logged fields.
- Channel drain.✓
 - Slope drain.✓
 - Herringbone.✓ (Any 1) (1)
- 6.4.3 A few aspects that can cause problems if there is no drainage system installed around the perimeter of a building.
- Water leaks into the house.✓
 - Moisture rises into the walls.✓
 - Standing water attracts pests.✓
 - Erosion around the foundation.✓ (Any 3) (3)
- 6.5 6.5.1 Identification of timer A.
- Mechanical timer.✓ (1)
- 6.5.2 TWO disadvantages of timer B.
- Difficult to set up/operate.✓
 - Battery needs to be replaced regularly.✓
 - Electronic parts need to be properly sealed.✓ (Any 2) (2)
- 6.5.3 The timer that has a multiple programming function.
- Timer B.✓ (1)
- 6.6 THREE advantages of using the faucet water filtration system.
- Does not require the boiling of water.✓
 - Quick process of purifying water.✓
 - Filters are easy accessible on the kitchen top.✓
 - Can be switched on and off.✓
 - Cartridges are fairly inexpensive and easy to change.✓ (Any 3) (3)
- 6.7 A monitoring system for each scenario in the table below.

Scenario	Monitoring system
Precisely control the rate of application of fertilizer.	6.7.1 Variable-rate Technology.✓
Determine areas of under growth.	6.7.2 Geographic Information System/Drone/Thermal imaging.Yield monitor.✓

(2)
[30]

TOTAL SECTION B: 160
GRAND TOTAL: 200